MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 11 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. How many terms of the AP $9, 17, 25, \ldots$ must be taken to give a sum of 636? [1]

(A) 11 (B) 12 (C) 13 (D) 14

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 12** [0.5 Mark]

Given: $a = 9$, $d = 17 - 9 = 8$, $S_n = 636$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

$\frac{n}{2}[18 + 8(n-1)] = 636 \implies n(4n+5) = 636 \implies 4n^2 + 5n - 636 = 0 \implies (n-12)(4n+53) = 0$.

Since $n \in \mathbb{N}$, $n = \mathbf{12}$.

Q2. In an AP, if $a = 17$, last term $l = 350$, and $d = 9$, then the number of terms n is: [1]

(A) 36 (B) 37 (C) 38 (D) 39

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 38** [0.5 Mark]

Given: $a = 17$, $l = 350$, $d = 9$.

Formula / Theorem Used: $l = a + (n-1)d$

Justification: [0.5 Mark]

$17 + 9(n-1) = 350 \implies 9(n-1) = 333 \implies n - 1 = 37 \implies n = \mathbf{38}$.

Q3. If the nth term of an AP is $a_n = 4n - 5$, then the sum of its first 25 terms (S_{25}) is: [1]

(A) 1175 (B) 1150 (C) 1200 (D) 1225

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 1175** [0.5 Mark]

Given: $a_1 = 4(1) - 5 = -1$, $a_{25} = 4(25) - 5 = 95$.

Formula / Theorem Used: $S_n = \frac{n}{2}(a_1 + a_n)$

Justification: [0.5 Mark]

$S_{25} = \frac{25}{2}(-1 + 95) = \frac{25}{2} \times 94 = 25 \times 47 = \mathbf{1175}$.

Q4. If $p-1$, $p+3$, and $3p-1$ are three consecutive terms of an AP, then p equals: [1]

(A) 4 (B) -4 (C) 2 (D) -2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 4** [0.5 Mark]

Applying the condition for consecutive AP terms a, b, c:

Formula / Theorem Used: $2b = a + c$

Justification: [0.5 Mark]

$2(p+3) = (p-1) + (3p-1) \implies 2p+6 = 4p-2 \implies 2p = 8 \implies p = \mathbf{4}$. (Terms: $3, 7, 11$.)

Q5. **Assertion (A):** The sequence $\sqrt{3}, \sqrt{12}, \sqrt{27}, \sqrt{48}, \ldots$ is an AP.
Reason (R): Simplifying the surds gives $\sqrt{3}, 2\sqrt{3}, 3\sqrt{3}, 4\sqrt{3}, \ldots$ with constant common difference $d = \sqrt{3}$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true, and (R) explains (A).** [0.5 Mark]

Checking successive differences:

Formula / Theorem Used: $a_2 - a_1 = 2\sqrt{3} - \sqrt{3} = \sqrt{3}, \quad a_3 - a_2 = 3\sqrt{3} - 2\sqrt{3} = \sqrt{3}$

Justification: [0.5 Mark]

Reason (R) reduces each radical accurately and verifies constant difference $\sqrt{3}$ (true), thereby proving Assertion (A).

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SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. Find the sum of the first 15 positive multiples of 8. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying the AP [1 Mark]

First 15 positive multiples of 8: $8, 16, 24, \ldots$ ($a = 8$, $d = 8$, $n = 15$).

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 2: Evaluating S_{15} [1 Mark]

$$S_{15} = \frac{15}{2}[2(8) + (15-1)(8)] = \frac{15}{2}(16 + 112) = \frac{15}{2} \times 128 = 15 \times 64 = \boxed{960}$$

Q7. Find the sum of the first 15 terms of an AP whose n^{th} term is given by $a_n = 9 - 5n$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding a_1 and a_{15} [1 Mark]

$a_1 = 9 - 5(1) = 4$ and $a_{15} = 9 - 5(15) = 9 - 75 = -66$.

Formula / Theorem Used: $S_n = \frac{n}{2}(a_1 + a_n)$

Step 2: Calculating S_{15} [1 Mark]

$$S_{15} = \frac{15}{2}(4 - 66) = \frac{15}{2} \times (-62) = 15 \times (-31) = \boxed{-465}$$

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. The sum of three numbers in AP is 21 and the product of the first and third numbers exceeds the second number by 6. Find the three numbers. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Represent the three numbers symmetrically as:

Formula / Theorem Used: Three consecutive terms in AP: $(a-d)$, a, $(a+d)$

Step 1: Finding the Middle Term a [1 Mark]

Sum $= 21 \implies (a-d) + a + (a+d) = 21 \implies 3a = 21 \implies a = 7$.

Step 2: Using the Product Condition for d [1.5 Marks]

$(a-d)(a+d) = a + 6 \implies (7-d)(7+d) = 7 + 6 \implies 49 - d^2 = 13 \implies d^2 = 36 \implies d = \pm 6$.

Step 3: Listing the Numbers [0.5 Mark]

For $d = 6$: $7 - 6 = 1$, 7, $7 + 6 = 13$. (For $d = -6$: $13, 7, 1$.)

Required numbers: **1**, **7**, **13** (or **13**, **7**, **1**)

Q9. Find the 4th term from the end of the AP $-11, -8, -5, \ldots, 49$. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying l and d [1.5 Marks]

Here first term $a = -11$, common difference $d = -8 - (-11) = -8 + 11 = 3$, and last term $l = 49$.

Formula / Theorem Used: n^{th} term from end $= l - (n-1)d$

Step 2: Evaluating 4th Term from the End [1.5 Marks]

$$4^{\text{th}} \text{ term from end} = 49 - (4-1)(3) = 49 - 3 \times 3 = 49 - 9 = \boxed{40}$$

Q10. A manufacturer of laptop batteries produced 600 units in the 3rd year and 700 units in the 7th year. Assuming that production increases uniformly by a fixed number every year, find: (a) production in the 1st year, (b) production in the 10th year, and (c) total production in the first 7 years. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Annual production forms an AP with first term a and uniform increase d. Given $a_3 = 600$ and $a_7 = 700$.

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

(a) Production in the 1st Year (a) [2 Marks]

$a + 2d = 600$ — (i) and $a + 6d = 700$ — (ii). Subtracting (i) from (ii):

$$4d = 700 - 600 = 100 \implies d = 25$$

Substituting $d = 25$ into (i): $a + 2(25) = 600 \implies a = 600 - 50 = \mathbf{550}$ units.

(b) Production in the 10th Year (a_{10}) [1.5 Marks]

$a_{10} = a + 9d = 550 + 9(25) = 550 + 225 = \mathbf{775}$ units.

(c) Total Production in First 7 Years (S_7) [1.5 Marks]

$S_7 = \frac{7}{2}(a_1 + a_7) = \frac{7}{2}(550 + 700) = \frac{7}{2} \times 1{,}250 = 7 \times 625 = \mathbf{4{,}375}$ units.

(a) **550** units, (b) **775** units, (c) **4,375** units

Q11. If the sum of the first p terms of an AP is q and the sum of its first q terms is p ($p \neq q$), prove that the sum of its first $(p + q)$ terms is $-(p + q)$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let first term be a and common difference be d. Given $S_p = q$ and $S_q = p$ with $p \neq q$.

Formula / Theorem Used: $S_k = \frac{k}{2}[2a + (k-1)d]$

Step 1: Writing Equations for $S_p = q$ and $S_q = p$ [1.5 Marks]

$$\frac{p}{2}[2a + (p-1)d] = q \implies 2ap + p(p-1)d = 2q \quad \text{— (i)}$$

$$\frac{q}{2}[2a + (q-1)d] = p \implies 2aq + q(q-1)d = 2p \quad \text{— (ii)}$$

Step 2: Subtracting (ii) from (i) and Dividing by $(p - q)$ [2 Marks]

$$2a(p-q) + [(p^2 - q^2) - (p-q)]d = 2(q-p) = -2(p-q)$$

Factoring out $(p - q)$:

$$(p-q)[2a + (p+q-1)d] = -2(p-q)$$

Since $p \neq q$, dividing by $(p - q)$ yields: $2a + (p + q - 1)d = -2$. — (iii)

Step 3: Evaluating S_{p+q} [1.5 Marks]

$$S_{p+q} = \frac{p+q}{2}[2a + (p+q-1)d] = \frac{p+q}{2} \times (-2) = \boxed{-(\mathbf{p+q}) \quad \text{[Hence Proved]}}$$

Q12. CASE STUDY: State Championship Archery Target Practice

Rhea prepares for the state archery championship over a 21-day intensive camp. She shoots 30 arrows on Day 1 and increases her daily target uniformly by 5 arrows every day. [5]

(i) How many arrows does Rhea shoot on the 6th day? [1]

(ii) On which day does she shoot exactly 100 arrows? [1]

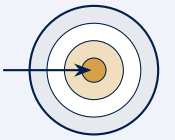
(iii) Calculate the total number of arrows Rhea shoots during the entire 21-day camp. [3]

— **OR** —

Find the ratio of arrows shot on the final (21st) day to those shot on Day 1. [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Daily arrow counts form the AP $30, 35, 40, \ldots$ with $a = 30$, $d = 5$, and $n = 21$.



Day 1: 30, +5/day

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

Sub-part (i): Arrows Shot on the 6th Day [1 Mark]

$a_6 = 30 + (6-1)(5) = 30 + 25 = \boxed{\textbf{55} \text{ arrows}}$.

Sub-part (ii): Day Reaching 100 Arrows [1 Mark]

$30 + (n-1)(5) = 100 \implies 5(n-1) = 70 \implies n - 1 = 14 \implies n = \boxed{\textbf{15}^{\text{th}} \text{ day}}$.

Sub-part (iii): Total Arrows Shot Across All 21 Days [3 Marks]

$$S_{21} = \frac{21}{2}[2(30) + (21-1)(5)] = \frac{21}{2}(60 + 100) = \frac{21}{2} \times 160 = 21 \times 80 = \boxed{\textbf{1,680} \text{ arrows}}$$

Sub-part (iii) [OR Option]: Ratio of Arrows on 21st Day to Day 1 [3 Marks]

$a_{21} = 30 + (21-1)(5) = 30 + 100 = 130$, and $a_1 = 30$.

$$\frac{a_{21}}{a_1} = \frac{130}{30} = \frac{13}{3} \implies \boxed{\mathbf{a_{21} : a_1 = 13 : 3}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★